Shreyas Pawar

Prompt Engineer | NLP & Speech Text Processing

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Portfolio | GitHub | GitHub (NeMo contributions): shrpawar-alt

Summary

Engineer with two years of experience in language-technology data and tooling, centred on open-source contributions to NVIDIA's NeMo-text-processing library, where I build and maintain Hindi text normalization (TTS front-end) grammars. Additional hands-on work in RLHF preference evaluation, supervised fine-tuning data design for code LLMs, and multimodal question answering. Seeking a Master's programme in NLP to deepen research in multilingual language processing and LLM alignment.

Experience

Zensar Technologies Oct 2024 – Present

Prompt Engineer *Remote*

Open-source: NVIDIA NeMo-text-processing, Hindi Text Normalization all PRs

- Design and maintain Hindi text-normalization grammars (weighted finite-state transducers in *pynini*, tagger–verbalizer architecture) used to prepare text for speech synthesis; Hindi TN v3 is merged to `main`, v4 is in active development, with review by the NeMo maintainers.
- Raised the date class accuracy from ~87% to ~99% (#418); built the Serial and Roman-numeral taggers (#440, #442) and extended Roman support to 3999 with a compositional design (#463).
- Reworked Electronic, Money and Measure handling: partial TSV matching, relative paths, singular/plural currency forms, and removal of unit/currency false positives (#424, #425, #429, #458).
- Optimised Address, Electronic, Serial and Cardinal grammars, including Indian lakh/crore comma grouping (#461); added percent and zero-padded time fixes, math expressions and negative-number verbalization (#483, #486, #505, under review).
- Practise TSV-driven, non-hardcoded, compositional grammar design, with regression and blind-set testing and Sparrowhawk deployment checks.

LLM data and evaluation

- Produced pairwise preference judgements for RLHF, comparing model responses on correctness, helpfulness and instruction following.
- Authored prompt–response pairs for supervised fine-tuning of code LLMs in Python and C++, and wrote rubrics to assess them.
- Answered visual question-answering tasks on technical diagrams, creating questions that require multi-step reasoning.

Projects

- **AI Chatbot on Discord**: conversational bot for Discord servers built on the Google Generative AI API.
- **Student Record Management System** and **Bank Management System**: C++ applications using object-oriented design with file-based record storage.

Education

Savitribai Phule Pune University 2024

B.E. Computer Engineering, PVG's College of Engineering & SSDIOM, Nashik *CGPA 8.92 / 10*

Technical Skills

- **NLP & speech text:** text normalization, weighted finite-state transducers (pynini / OpenFST), Sparrowhawk, tokenization, Hindi and code-mixed text
- **LLM work:** prompt engineering, RLHF preference data, SFT data creation, response evaluation, multimodal QA
- **Languages & tools:** Python, C++, Git and GitHub (PR workflow), Linux/WSL, LaTeX

Research Interests

Multilingual and low-resource NLP, speech text front-ends, LLM alignment and evaluation.